



Enhancing sign language recognition using CNN and SIFT: A case study on Pakistan sign language

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ABSTRACT

Millions of people who have trouble hearing rely heavily on sign language as their primary means of communication. As a form of visual language, sign language is primarily used as features by the deaf and hard of hearing to communicate with one another. These features make it challenging for the average person to grasp what is being said. As a result, it becomes more difficult for deaf people to interact with the hearing community. In order to solve this communication barrier, scientists have been working on sign language recognition systems. The purpose of this research is to propose a hand gesture recognition framework for Pakistan Sign Language (PSL) by training a Convolutional Neural Network (CNN) on PSL gesture images obtained with the aid of a testbed built in the laboratory using the Kinect motion sensor for Urdu alphabets. Kinect images of hands were taken in this study under a variety of lighting conditions. A feature vector set was created from hand margin, size, shape, palm center points, and finger position using Scale Invariant Feature Transform (SIFT). Important features of PSL-based sign images were extracted using SIFT and then converted to vector points. The proposed improved CNN model was able to achieve an impressive accuracy of 98.74% on the PSL dataset, demonstrating an impressively low error rate of only 1.26%. A case study is demonstrated in the assessment of the system that has been provided. Finally, a comparison is performed between others and our work to show the efficiency of the proposed framework and legitimize the concept.

1. Introduction

Body language and other non-verbal cues are referred to as “gestures” when they are used to convey meaning. The use of gestures is commonplace and widely recognized as an effective means of communication. The use of hand gestures in everyday life is so ingrained that many people continue to do so even when using mobile phones (Goyal et al., 2013). Human-Computer Interaction (HCI) has made extensive use of gesture recognition, opening up the world of communication to the hearing-impaired and the speech-impaired. This highlights the importance of developing automatic gesture recognition systems. As a result of these factors, there is a greater demand for systems that are capable of recognizing hand gestures automatically. First, the global population of the deaf and mute is growing. About 430 million people worldwide have trouble hearing or communicating, according to the

World Health Organization (WHO) (Goyal et al., 2013). The WHO estimates that 5 percent of the population worldwide has some form of hearing loss. Second, video games and Virtual Reality (VR) based software are increasingly utilizing vision-based and touchless applications. Sign language is one of the most significant applications of hand gestures. With its usefulness in facilitating communication, hand gesture provides an inspiring area in which to conduct the study.

All over the world, deaf communities rely on sign language as a means of communication. It combines both manual and automated processes. Signs that incorporate manual components are those in which two or more hand gestures are combined. Movement of the body and expressions of the face are examples of non-manual components of gesture (Javaid et al., 2022). Traditional sign languages make use of both static and dynamic gestures. Images of hands making static gestures don't show any motion on the part of the hands. As a result, it can

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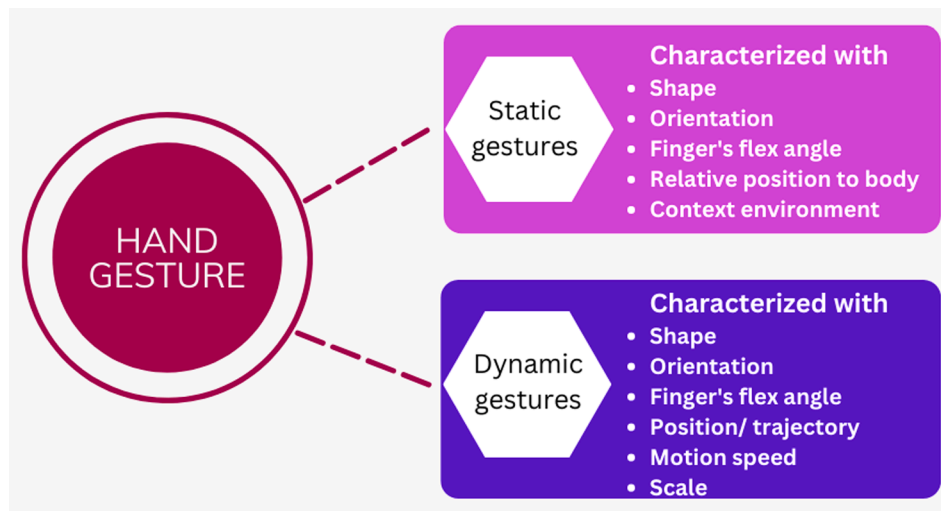


Fig. 1. Classification of Hand gesture, Reproduced (Pisharady and Saerbeck, 2015 Dec).

be concluded that the hands remain in the same position. Dynamic gestures, on the other hand, are created by the movement of the hands and arms. Making and interpreting dynamic gestures requires more effort. Most people can decipher these motions by looking at a series of pictures from beginning to end (Haroon et al., 2022 Sep 30). Classifying hand gestures into static and dynamic categories is depicted in Fig. 1.

Because most people do not know or understand sign language, there is a barrier to communication between the deaf and hearing. Sign language is typically only understood by other experts or by deaf people themselves (Raees et al., 2016). Because many hearing-impaired people do not know sign languages, this communication barrier affects the majority of deaf and mute people as well. Because of the rise in listening and hearing impairments around the world, researchers and practitioners have become increasingly interested in finding solutions to problems in sign language and developing highly satisfying answers (Rastgoo et al., 2021).

There are two approaches that can bridge the gap in communication. The first step in communicating with the deaf community is to persuade a hearing person to learn sign language. A second option is to create a system that is capable of precisely converting the signs into a common form of human communication. It is difficult, if not impossible, to persuade a typical person to study sign language unless that person has a compelling reason to do so. There is a greater sense of durability and practicality with the second choice (Amin et al., 2022). Researchers and academics from all over the world are constantly making progress on it. Every country has its unique way of using sign language; there is no standardized international sign language. In the same way that American Sign Language (ASL), Australian Sign Language (Auslan), and British Sign Language (BSL) are all distinct varieties of sign language, so is American Sign. Similarly, to spoken language, Pakistan also has its own sign language, known as Pakistan Sign Language (PSL). This language employs the Urdu script “اردو” (Cheok et al., 2019), the script used for the national language of Pakistan. The grammar, vocabulary, and syntax of PSL are distinguishable from those of other sign languages.

One of the many challenges in the identification field that gesture recognition must overcome is the inability to properly identify sign language hand gestures. Some researchers have focused on ASL, but not nearly as much has been done to improve PSL recognition (Shah et al., 2021). There is also the problem of gestures in PSL having a similar context, which is a challenge for researchers. Some PSL gestures have very little variation in their posture from one another. These gestures can be distinguished from one another by the subtle differences in the thumb and fingers. Although well-intentioned, non-native sign language speakers may cause confusion when using gestures intended for sign

language users. It's not easy to figure out how to recognize these gestures.

Hand gesture recognition researchers have used two different approaches: sensor-based and computer vision. Devices such as gloves equipped with sensors allow for the recognition of hand gestures. It was necessary to wear these sensor gloves on one's hands. The motion of the hand and fingers can be taken as a representation of a bodily reaction. This response was utilized in the process of data collection. After data collection, a computer wired to the gloves processed the information. These sensor gloves record actions such as recording the position of the hand and fingers (Ahuja and Singh, 2016).

With the development of glove sensors, HCI for hand gestures was officially launched. It's a straightforward interface for issuing commands to computers. Sensors built into gloves can track the wearer's hand in space and time by identifying the wearer's precise finger and palm coordinates. Image noise is a major detractor from the reliability of computer vision-based gesture recognition systems. The recognition procedure is affected in various ways by the available light. In real-time scenarios, recognition becomes more challenging when the lighting suddenly changes. While much progress has been made in gesture recognition under a single lighting source, there is still considerable room for improvement when working under multiple sources. There are a number of issues that prevent automated systems from providing reliable and effective recognition of Urdu Sign Language (USL). Here are some of them (Haroon et al., 2022 Sep 30)- (Shah et al., 2021):

- Sign language data that is rich in variety and annotation is essential for training accurate recognition systems. Due to a lack of publicly accessible annotated sign language data, however, it is difficult to collect an extensive set of data for USL. Because of this lack of information, models built on USL data may not be able to generalize well or recognize all possible expressions and gestures.
- Background objects with a similar color to human skin pose a challenge for hand segmentation algorithms' accuracy.
- When attempting to recognize a gesture, it's possible that mistakes will be made if the gesture has a similar enough shape and orientation to another.
- Different regions use slightly different forms of sign language. Due to these distinctions, it is challenging to create a unified recognition system capable of recognizing signs in all USL dialects and regions.
- There is a strong emphasis on nonverbal cues like gestures and body language when using sign language. However, it is more difficult to recognize these non-manual signs than it is to recognize manual ones.

The following is a list of the major contributions that this study makes:

- Proposed an efficient hand gesture-based recognition framework for PSL that works well with various lighting conditions.
- The dataset for PSL hand gesture images is collected under different lighting conditions for Urdu alphabets;
- SIFT method is used for significant feature extraction for hand gesture recognition.
- Segmentation and identification of key points within features to minimize the time taken for localization of substantial points.
- Finally, to prove the validity of the concept and the efficacy of the proposed framework, we compare it to previous research.

The structure of the rest of the paper is organized as follows. [Section 2](#) gives detailed information on relevant literature reviews. [Section 3](#) explains the proposed framework for hand gesture recognition of PSL. Experimental results and discussion are presented in [Section 4](#). [Section 5](#) contributes to the conclusion and possible future work.

2. Literature Review

Sign language gesture recognition is a hot topic in the field of artificial intelligence. Previous research into gesture recognition has made extensive use of sign languages. Colored hands, colored markers, and gloves were among the gesture recognition methods proposed by the researchers. As stated in ([Raees et al., 2016](#)), a method was created to recognize numerical and alphabetical characters in PSL. The proposed system successfully identified all digits 0 through 9 and all alphabets with the thumb visible. The position of the thumb was identified using template matching, and the fingers were identified using deep pixel-based analysis. The algorithm consisted of a seven-stage process that was carried out to achieve recognition. During the first stage, the hand posture is determined by analyzing the image. The RGB mode was used to capture the images from the live streams. Cut and mined RGB hand images were done before moving on to the next phase. During the second stage, they identified the edges. To achieve this goal, the Sobel operator was opted for. Thumb and finger location are detected in stages three and four via the matching of templates and extensive pixel-based analysis. The algorithmic fingerprint recognition process utilized a sliding method. First, the detection of all fingers was accomplished, and then individual fingers were classified. As a means of making the system user-agnostic, the signers' upper-finger width was measured. The finger's location is used to determine which finger it is, from the smallest finger on the left hand to the largest on the right. The fifth stage involved identifying groups by using the thumb and index finger as markers. Groups of letters were designated as positions G1 through G3, and groups of numbers, G4 through G7. This classification is based on how far apart the thumb is from the other fingers. The sixth and seventh steps, respectively, are sign recognition and output. The computer was fed a series of gestures. It gives each finger a different name and identifier. It could tell which finger was used to make a sign. The OpenCv library and Visual Studio 2010 were used to construct the model. The system had a straightforward layout and buttons that were simple to use. By simply clicking the "capture" button, signs could be easily recorded. Five individual testers tested the system to ensure its stability. Each sign was tested 20 times to ensure the accuracy of the proposed system. For gestures with an orientation of the hands greater than ten degrees, the proposed system incorrectly identified them. Images of signs without a thumb orientation were also unrecognized. The proposed system achieved 84 % accuracy with 16 % system failure thanks to its light insensitivity, low cost, and signer independence.

One-handed static sign recognition using PSL video and a Support Vector Machine (SVM) trained with multiple kernels ([Shah et al., 2021](#)). This was a no-frills approach that could be executed without any special equipment. For this categorization, researchers used a whopping thirty-

six different alphabets! Three different kernel functions (Linear, Gaussian, and polynomial) were used to determine which one was most effective for the sign language detection problem, using the multiple kernel learning idea. Frames were captured from PSL videos and six signers were recorded to create a dataset for classification. The camera, background, and clothing color were all kept constant during photo-shoots that were otherwise staged. Images captured were environment-dependent due to deviations in hand angle, lighting, and hand scaling. A typical signing setup places the camera anywhere from 1 to 2 feet away from the signer. With a massive dataset including 6633 images of one-hand signs, 184 image samples of each alphabet were taken, on average. The camera was used to capture images of hands as input, and these images were then segmented using a K-means clustering approach. Grayscale image conversion followed segmentation for feature extraction. An automated cropping procedure was used to get rid of the extra space. Feature extraction is given grayscale images as input. Histogram of Oriented Gradients (HOG), Edge Orientation Histogram (EOH), Local Binary Patterns (LBP), and SURF were the four features taken from the original image. Classification of a signal is performed using the kernel that was selected for every attribute using the kernel selection algorithm. The accuracy of three different kernel functions was compared to pick the best one. In a predictive sense, the proposed system was 91.98 percent accurate. With an overall accuracy of only 15 %, the SURF feature was useless for sign language recognition.

The authors ([Mirza et al., 2022](#)) made progress in static and dynamic PSL sign recognition utilizing two distinct Bag-of-Words (BoW) and SVM techniques. The primary goal of this study was to amass a dataset and create a visual system for PSL recognition. The dataset included 5120 still images for use in static signage and 353 moving images for use in dynamic signage. Ten native PSL signers helped with the collection of these images and videos. Both data collection and analysis were subsectioned as their methodology sections for this study. Over a three-month period, native signers provided their expertise during the data collection stage. In total, 36 still images and 3 moving videos of signs have been taken with the support of signers. Applying the recognition methods to the data is what they did during the analysis stage. Pre-processing, segmentation, feature extraction, and classification were all parts of the process. Image and video data collected were organized and stored in folders with descriptive names. Because it's more difficult to process moving signs than still ones, the researchers processed the videos frame by frame so that they appeared to be still photographed. Resizing the captured images helped reduce computational complexity and time. Grayscale versions of the images were created after they were resized. Hand segmentation from these images was accomplished using the bounding box technique. The threshold was kept low in order to extract the entire hand from the picture. After image feature extraction, SURF was selected. There were 4096 training images and 1024 test images. By combining BoW and SVM methods, this research created a vision-based system for PSL sign recognition. The model that makes classification predictions based on input images was trained using a support vector machine classifier. 500 bags were used for static signs, and the result was a success rate of 97.80 percent. Two hundred bags were used for the dynamic signs. Fifteen models were trained on video frame data for dynamic sign recognition. The 96.53 % success rate for flexible signs is the highest success rate for flexible signs in PSL.

Researchers have been working on a Convolutional Neural Network (CNN)-based real-time recognition system for Arabic Sign Language (ArSL) ([Alsaadi et al., 2022](#)). The authors also brought up the problem of not everyone having access to hardware devices such as color-based gloves. The feature-extracting model of machine learning was the foundation for the vast majority of the works by various researchers. Extracted features have a significant impact on the outcome of these projects. The authors proposed a convolutional neural network (CNN) based model that, unlike machine learning techniques, extracts the features on its own. The proposed structure is broken down into a number of stages. Dataset selection was the initial process. The data used

in this research was obtained from the general public. A total of 54,049 grayscale images of the 32 ArSL alphabets were included in the dataset. The second step involved choosing a suitable deep learning architecture; the Convolutional Neural Network (CNN) is currently the most widely used algorithm for recognition problems. They resized and reorganized the dataset so that it could be used with four distinct CNN designs. Python library labels were used to separate the dataset into a training set and a test set. Training data comprised 80 % of the total, while test data constituted 20 %. The learning rate was managed with the help of Adam's optimizer. Background and hand position played a significant role in the system's performance. The system's efficiency suffered after a change in setting.

A novel machine translation model that used natural language processing (NLP) techniques to convert English sentences into PSL was proposed by (Khan et al., 2020 Jul). The purpose of this study was to overcome the challenges faced by researchers. These challenges include the absence of proper grammatical sentence structure for converting sentences into PSL and proper written grammatical rules. The dataset contains the collection of English sentences and categorization based on tenses and sentence structure. Grammatical rules were formulated and applied for the translation of PSL sentences. These sentences were translated into PSL signs with the help of deaf participants. A minimum of three participants were involved in one sign translation to capture different variations of a word. Here grammatical rules were developed for PSL sentence structure and converted into context-free grammar syntax. This study was implemented using NLP techniques. An avatar-based software system was developed to translate the words into gestures. This avatar took the sentences as input and translated them into sign language. The proposed system evaluation was done both automatically and manually. The evaluation showed that the system Bilingual Evaluation Understudy (BLEU) scored 0.78. The proposed system performed very well with simple sentence translation but was unable to translate sentences with compound and compound-complex structures.

Researchers have difficulty recognizing PSL in part because there are not enough publicly available datasets to analyze. The research reported in (Imran et al., 2021) aimed to produce a dataset and make it available for others to use. In this case, a webcam was used to capture the data, which was then released to the public. Creating a dataset, learning to recognize signs, and building an Android app were the three pillars on which their research rested. There are 37 different alphabets in the Urdu language; the first step in creating the dataset was to capture 40 images of each of these. Using a webcam, they were able to take pictures while rotating their fingers and hands in various ways. There were 37 different alphabets represented in 1480 images, each of which was categorized into one of the 37 different alphabet types. Segmenting the hand from the background is the first step in the sign recognition process. Segmentation produced a blob list that the detection phase then used to adjust. The Android app they created for hand recognition used the machine-learning algorithm Support Vector Machine (SVM) and received input in the form of photographs taken by the device's camera. Display the Urdu translation of the interpreted sign using the specially trained SVM model.

Recent research includes (Al-Hammadi et al., 2020), which makes use of hand gesture recognition to delve into the dynamic signals of PSL. More and more touchless applications, as well as an ever-increasing population of people who are deaf or hard of hearing, have elevated the importance of gesture recognition. The purpose of this research was to suggest a C3D-based framework for gesture recognition. The study of SSL (Saudi Sign Language) has been completed. All of the data came from phrases and words found in the SSL. Five recordings were made with a total of 40 people, some of whom were hearing impaired and some of whom were not. There were no limitations on the environment, lighting, or clothing of the participants while collecting data. Multiple deep-learning methods were used to recognize 40 dynamic hand gestures performed by 40 unique people in an unregulated setting. Multiple algorithms were used to standardize each RGB-converted frame of the

Table 1

Comparative Evaluation of Different Techniques across Varied Illumination Conditions.

Paper	Gesture Count	Technique	Background	Luminosity Shift
(Shah et al., 2021)	36	SVM	Heterogeneous	2-variant
(Mirza et al., 2022)	36(static) 3 (dynamic)	SVM	black	1-variant
(Imran et al., 2021)	37	SVM	Consistent	1-variant
(Saleem et al., 2023)	33	CNN	Noise	variant
(Shah et al., 2023)	36	MCSVM	complex	indoor
(Chua et al., 2022 May)	10	YOLOv3	clustered	N/A
(Khan et al., 2022)	12	CNN	Uniform	1-variant

input video. The algorithm described in (Viola and Jones, 2001) was used to pinpoint the signer's face and the rest of the body gestures. The 3DCNN was used to learn each gesture's unique feature in great detail. The features were classified using a multi-layer perceptron (MLP) and then globalized using an autoencoder. The overall performance was satisfactory, with some misclassification occurring due to ambiguous gestures (signs that look alike). This was because of the poor lighting as well as the blurring problem. Table 1 provides a detailed summary of several recent studies, including their respective contributions, Gesture Count, techniques, results, limitations, and Luminosity Shift ratio as follows:

3. Materials and methods

This section of the paper proposes a framework for the use of image processing in the recognition of PSL gestures. The primary goal of gesture recognition research is to develop a system capable of recognizing and comprehending a wide variety of human gestures for efficient communication.

3.1. Proposed framework for PSL

The proposed framework is comprised of four stages. The first step involves acquiring image data for Urdu language alphabets. The second step consists of two tasks: preprocessing the image data and performing hand segmentation on the image. In the third step, SIFT algorithm and CNN model are utilized for extracting features and training the model. Subsequently, in the fourth step, the trained model is applied to classify the image data. Fig. 2 demonstrates the block diagram of the proposed framework consisting of different phases is given by:

3.1.1. Dataset acquisition

The Urdu language comprises a total of 38 alphabets, but typically we observe 37 distinct hand gestures corresponding to the most significant Urdu alphabet (called in Urdu language حروف تہجی). The exclusion of one sign is due to its representation using a series of varied hand shapes (dynamic gesture) (Shah et al., 2023). However, our focus lies solely on studying signs that possess a static nature. These alphabets along with their respective signs are demonstrated in Fig. 3. The dataset was captured by using an operational testbed using Kinect Sensor.

3.1.2. Pre-processing and segmentation

One of the most important steps in the hand gesture recognition problem is the pre-processing and segmentation of hand from the input image. The basic purpose of segmentation is to separate signer's hand from the background. The segmentation of colored images is a difficult

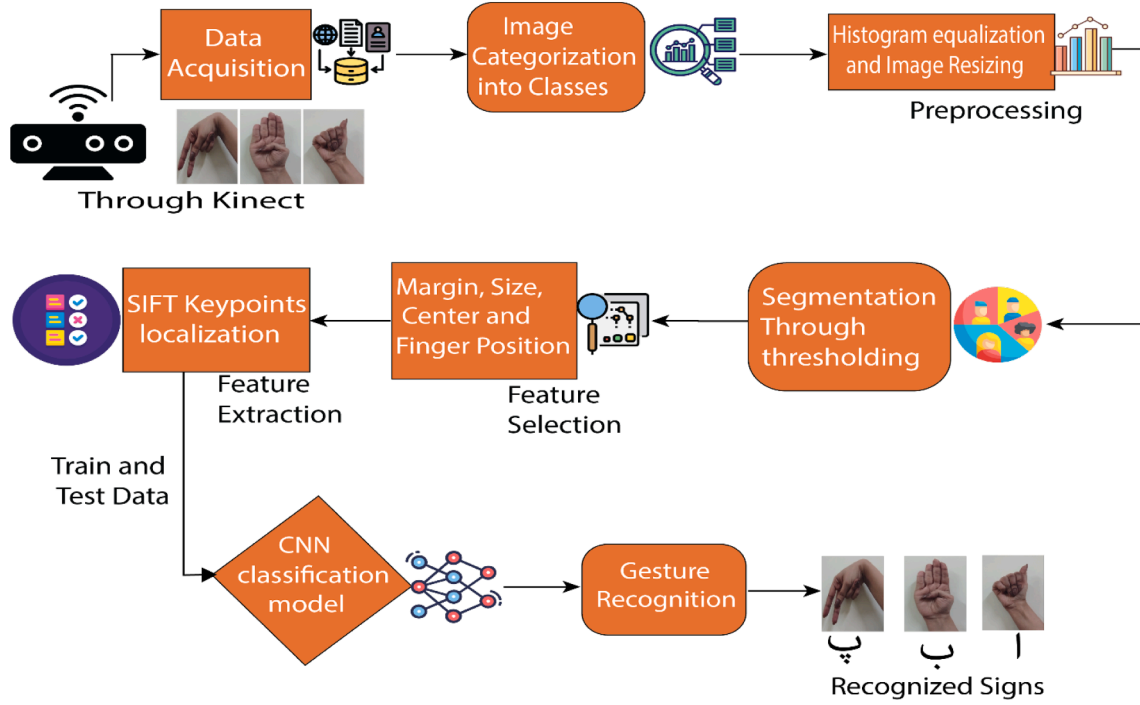


Fig. 2. Proposed framework for gesture recognition system for Pakistan sign language.

task because it is hard to find the correct thresholding values for light sensitivity and color saturation. We will convert the colored input images of the hand into grayscale for simplification. Grayscale images simplify algorithms as well as remove the complex requirements related to computations. Grayscale conversion method known as the luminosity method is used here (Haroon et al., 2022 Sep 30). Equation (1) shows the calculation method for the conversion of RGB images into grayscale.

$$\text{Grayscale} = 0.299R + 0.587G + 0.114B \quad (1)$$

Where R represents red color, G represents green, and B denote blue color.

Furthermore, a preprocessing technique of histogram equalization is applied to the images to enhance contrast values (Haroon et al., 2022 Sep 30). Following method can be used for calculating the histogram equalized image H shown in equation (2).

$$H(i, j) = \text{round}((I - 1) \sum_{n=0}^{g_{ij}} P_n) \quad (2)$$

Where "round" is a function that rounds down the value to the nearest integer. " I " is the number of possible intensity values usually 256. g represents the input image. Here P denotes the normalized histogram and n shows the values ranging from 0, 1... $I-1$. Segmentation is further done on the image data. This can be achieved by thresholding the images to identify the pixels that make hand region. After identifying the hand region, it is cropped from the remaining arm region of the images (Amin et al., 2022). The probability of a pixel value is given by the following equation (3):

$$P(x, y) = \begin{cases} 0 & \text{if } I(x, y) \leq T(x, y) \\ 1 & \text{if } I(x, y) > T(x, y) \end{cases} \quad (3)$$

3.1.3. Feature selection and extraction

The subsequent step in the recognition process is the extraction and selection of relevant features. Feature extraction is a critical step in the recognition process since irrelevant features do not contribute much to the recognition accuracy. Using fewer features can affect the accuracy and precision of the recognition process while using too many features

can increase the complexity and space requirements of the system. Therefore, feature extraction is a crucial step that requires careful consideration to ensure optimal performance of the sign language recognition system. Therefore, we used SIFT to extract relevant features from the obtained dataset (Naseem et al., 2019). Four important features (Margin or edge, size of hand, palm center, finger position) are extracted from SIFT. These feature points are illustrated in Fig. 4.

The Canny edge detection technique is utilized to detect the edges and boundaries of the human hand for computing the margin value. The edge detection method operates by identifying locations in an image where there is a variation in the intensity of pixel values. It functions by detecting discontinuity in the brightness of an image. Edge detection is accomplished by mining the magnitude and direction of the edges in image data. To calculate gradient, we will calculate magnitude and for orientation, direction is calculated. To calculate the orientation, entire image is divided into smaller portions, and for each portion, gradient and orientation are calculated. Gradients are the small changes in the x and y pixels of an image. For each image, the gradient of each pixel can be calculated by equations (4) and (5) (Cheok et al., 2019):

$$\text{Grad}_x(\text{row}, \text{col}) = I(\text{row}, \text{col} + 1) - I(\text{row}, \text{col} - 1) \quad (4)$$

$$\text{Grad}_y(\text{row}, \text{col}) = I(\text{row} + 1, \text{col}) - I(\text{row} - 1, \text{col}) \quad (5)$$

Where, Grad_x is the horizontal gradient and Grad_y is the vertical gradient. I is the input image. Now we will determine the magnitude and direction of each pixel value from the given equations (6) and (7) (Cheok et al., 2019).

$$M = \sqrt{(\text{Grad}_x)^2 + (\text{Grad}_y)^2} \quad (6)$$

$$\theta = \tan^{-1}(\text{Grad}_y / \text{Grad}_x) \quad (7)$$

Where, M represents gradient magnitude and θ represents angle of image pixels. The magnitude represents the strength of the gradient at each pixel, which is an indication of how rapidly the intensity changes in that direction. Hand size is an important feature for hand recognition because it provides information about the scale of the hand and can help

Sign	Sign Name	Alphabet	Sign	Sign Name	Alphabet	Sign	Sign Name	Alphabet	Sign	Sign Name	Alphabet
	Tay	ت		Pay	پ		Bay	ب		Aliph	ا
	Chay	چ		Jeem	ج		Thay	ث		Tey	ٹ
	Daal	ڈ		Daal	د		Khay	خ		Hey	ح
	Zay	ز		aRay	ڑ		Rey	ر		Dhaal	ذ
	Suaad	ص		Sheen	ش		Seen	س		Djay	ژ
	Ain	ع		Zoayn	ظ		Toayn	ط		Dhuad	ض
	Kaaf	ک		Quaaf	ق		Fay	ف		Ghain	غ
	Noon	ن		Meem	م		Laam	ل		Gaaf	گ
	Chhoti yeh	ی		Hamza	ء		Hay	ہ		Vao	و
										Bari yeh	ے

Fig. 3. PSL alphabets and their corresponding hand pattern configuration.

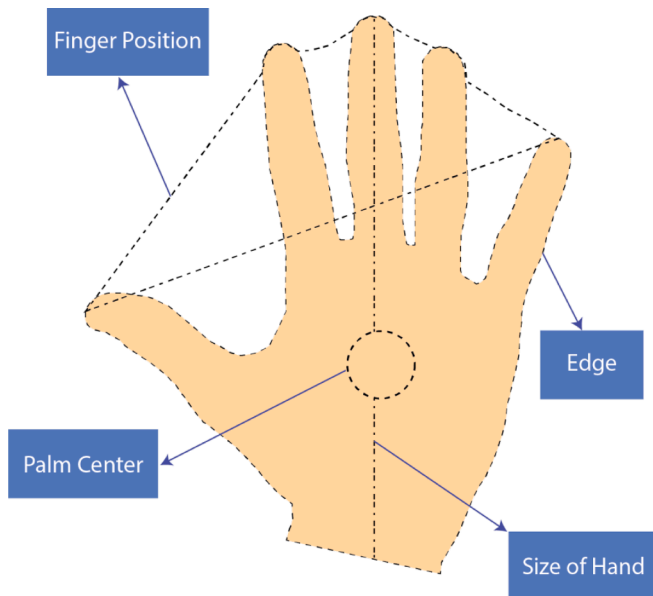


Fig. 4. The hand's selected attributes for constructing a feature set.

distinguish between hands of different sizes. To calculate the size of the hand the number of pixels in the hand region is utilized. We used the bounding box technique, which can be calculated using the minimum and maximum values of the x and y coordinates of the hand pixels. The hand size can then be calculated using the area of the bounding box or the number of pixels inside the bounding box (Cheok et al., 2019).

$$M_i = x_{\max} - x_{\min} \quad (8)$$

$$N_i = y_{\max} - y_{\min} \quad (9)$$

Where, $x_{\max}$, $x_{\min}$, $y_{\max}$ and $y_{\min}$ are the minimum and maximum xy -coordinates of the image and M_i and N_i are the width and height of the hand's bounding box, respectively (Cheok et al., 2019).

$$A_z = M_i(x, y) * N_i(x, y) \quad (10)$$

Where, A_z is the size of the hand and ' x ' and ' y ' are the coordinates of the hand pixels in the image.

Calculating the center of the palm feature is an important step in hand gesture recognition because it provides important information about the location and position of the hand in the image. Following function can be used for calculating the palm center points.

$$C_x = \frac{1}{A_z} \sum_{i=1}^n (x_i) \quad (11)$$

$$C_y = \frac{1}{A_z} \sum_{i=1}^n (y_i) \quad (12)$$



Fig. 6. Testbed for Kinect-based PSL dataset acquisition.

Where, C_x and C_y are the center points for x and y -coordinates respectively. To calculate the position of fingers using pixel coordinates, we can calculate the distance and angle between the fingertips and center of the hand by the following equations:

$$D(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (13)$$

$$\theta(x, y) = \tan^{-1} \sum_{i=1}^n ((x_i - y_i) * \frac{180}{\pi}) \quad (14)$$

where D is the distance of fingers and θ is the angle between fingers. Once the features have been extracted, they are combined into a feature vector set F_v . This feature set is given in equation (15) below.

$$F_v = [\text{Margin, size of hand, palm center points, finger position}] \quad (15)$$

Where,

$$\text{Margin} = F_1 = [m1, m2, m3, \dots, mn] \quad (16)$$

$$\text{Hand Size} = F_2 = [h1, h2, h3, \dots, hn] \quad (17)$$

$$\text{Palm center points} = F_3 = [p1, p2, p3, \dots, pn] \quad (18)$$

$$\text{Finger Position} = F_4 = [g1, g2, g3, \dots, gn] \quad (19)$$

Where, the total number of extracted feature points for each defined feature (F_1, F_2, F_3 and F_4) for a hand gesture is denoted by the letter " n ". The resulting matrix set after combining all the features is shown in equation (20).

$$S = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix} = \begin{bmatrix} m1, m2, m3, \dots, mn \\ h1, h2, h3, \dots, hn \\ p1, p2, p3, \dots, pn \\ g1, g2, g3, \dots, gn \end{bmatrix} \quad (20)$$

PSL-based images with relevant features under various lighting circumstances are analyzed with the SIFT method for feature extraction

and key point detection. Following is Algorithm 1 describing the SIFT algorithm.

Algorithm 1: Proposed SIFT algorithm for calculating key-points from PSL images

Input: Image I
Output: Image keypoints
 Read the input image I .
 Calculate d by using difference of gaussian function (DoG).
 Calculate initial keypoints (pd, sd, kd) of (DoG).
 Remove keypoints below some threshold (p, s, k).
 Eliminating unstable and inaccurately localized keypoints (p, s).
 Assigning orientation to each keypoint (p, s, θ).
 Generate feature descriptor by scale-space gradient (us, vs).
 List of described keypoints after descriptor generator (p, s, θ, u, v).

3.1.4. Convolutional Neural network

There are three types of layers in a convolutional neural network (CNN): the input layer, hidden layer and output layer. Each hidden layer further include convolution layers, the pooling layer, and the fully-connected layer. Each convolution layer is followed by a matching pooling layer. These convolutional and pooling layers have feature maps that are laid out in a 2D structure (Jadooki et al., 2017). Extracting spatial dependencies from the input data, convolutional layers pull out local features and patterns. Sample feature maps are reduced in dimension and gain translation invariance when pooled. When multiple features have been learned, they can be combined in a fully connected layer to make more complex predictions and classifications. These networks allow for the creation of deep network architectures by stacking multiple layers (Oyedotun and Khashman, 2017 Dec).

Algorithm 2: Proposed CNN Algorithm for classification

Input: List of keypoints (p, s, θ, u, v).
Output: Prediction results
 for ($r = 0; r < R; r++$) {
 for ($t = 0; t < T; t++$) {
 for ($j = 0; j < J; j++$) {
 amount = bias[r]
 for ($y = 0; y < Y; y++$) {
 for ($q1 = 0; q1 < Q1; q1++$) {
 for ($q2 = 0; q2 < Q2; q2++$) {
 input_value = input(y)(t + q1)(j + q2)
 weight_value = weight(y*r)(q1*q2)
 amount += weight_value*input_value
 }
 }
 }
 }
 }
 }

(continued on next page)



Fig. 7. Gestures captured at different illumination conditions (a) moderate light (b) bright light (c) dim light.

(continued)

Algorithm 2: Proposed CNN Algorithm for classification

```

}}}
output(r)(t)(j) = activation_func(amount)
}}}
for (a = 0; a < t; a++) {
    temp = 0
    for (b = 0; b < j; b++) {
        temp += W[i][j] * X[b]
    }
    Z[a] = temp + bias[a]
}

```

4. Experimentation and results

The experimentation phase starts with establishing a functional test environment within the laboratory to conduct the necessary experiments. This setup utilizes the Microsoft Kinect sensor to capture images and subsequently transform them into depth images, enabling the acquisition of data with varying resolutions. This setup provides the ability to obtain a comprehensive range of visual information, crucial for

the accurate analysis and evaluation of the proposed approach. Fig. 6 shows the testbed used in the experimental setup. It illustrates the arrangement of the Microsoft Kinect sensor and its positioning in relation to the participant. The Kinect sensor's ability to capture both RGB and depth images allows a comprehensive processing and analysis. This testbed serves as a crucial component in the experimental phase, enabling the collection of relevant data and ensuring the reliability and validity of the conducted experiments.

The next step implies the process of capturing the dataset. Twenty people helped with the information gathering process to generate the PSL datasets. Each participant performed signs under three lighting conditions (bright, moderate and dim/dull). A variety of participants were used for dataset acquisition process. Out of these 20 people, 10 female and 10 male participants were involved in the data acquisition phase. The age of the participants ranged between 20 and 40 years old. With 20 participants we can potentially include a diverse range of individuals in terms of age, gender, background, and proficiency in sign language. This diversity contributes to a more representative sample, leading to findings that may generalize better to broader populations (Khan et al., 2022 Jun 30). To capture the gesture images, participants

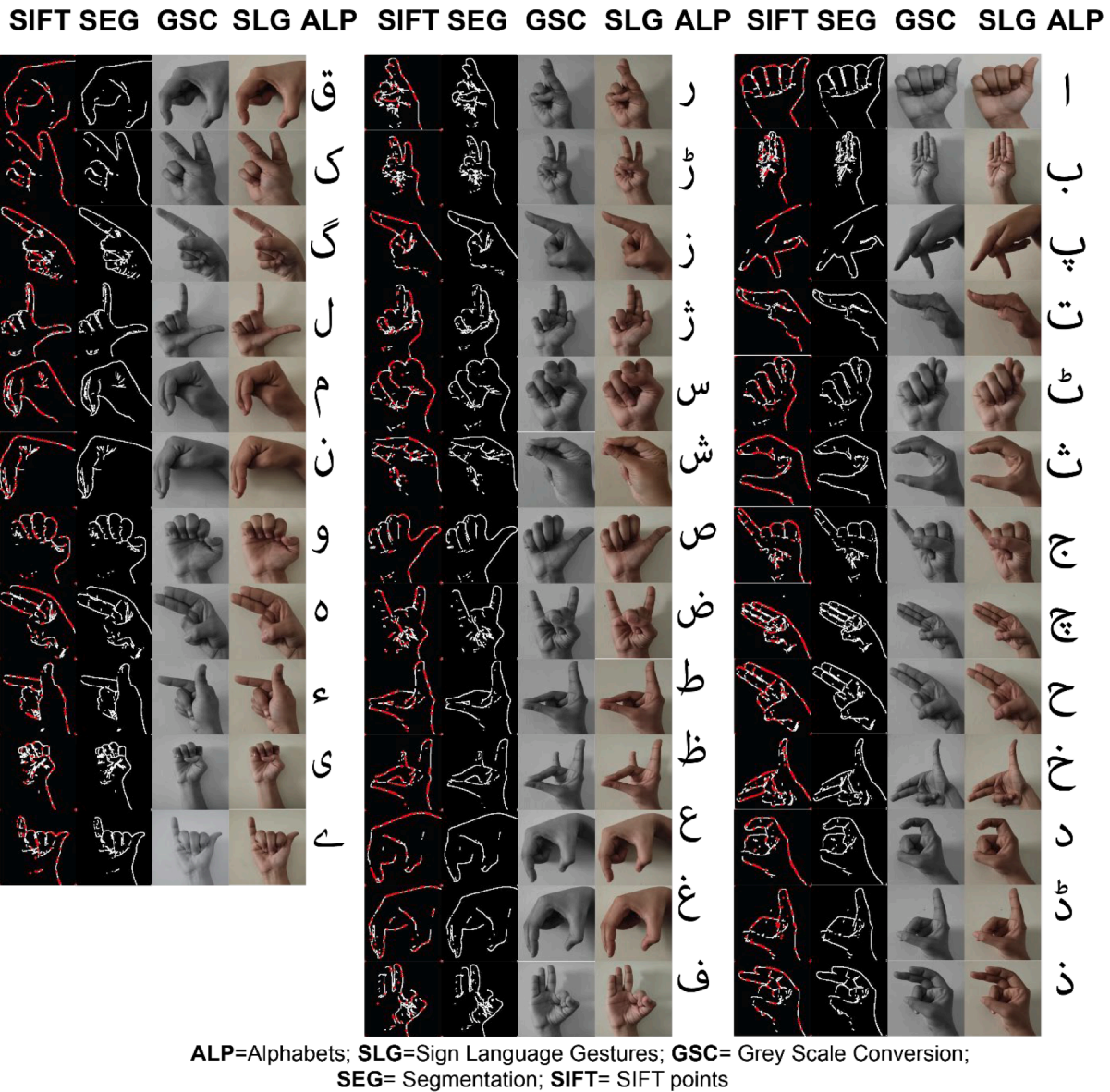


Fig. 8. PSL gestures and their conversion into grayscale and segmentation.

were placed at three different distances, 2ft, 3ft and 4ft from the testbed. A total of 2220 images were captured with the help of 20 participants. The captured gesture images were saved in PNG format. To create training and testing dataset, it is common practice to divide them from a single dataset. Images of captured dataset with different lighting conditions are shown in Fig. 7 below. The dataset was split into two parts: a training dataset and a testing dataset. About 70 % of the data was used for training the model, while the remaining 30 % was reserved for testing the model's performance.

The subsequent step involves transforming the PSL images to shades of gray to facilitate segmentation, as described in Equation (1). A Grayscale image makes it easier to do segmentation. The segmentation process endeavors to differentiate the hand object from its ambient surrounding. Four features (Margin, size and shape of hand, palm center points, finger position) are extracted from the segmented images by using SIFT feature extraction algorithm. To convert the identified feature points into meaningful vector points, a feature descriptor method is employed. The Matlab tool is utilized to extract the necessary feature point values and transform them into significant points using the SIFT algorithm. This process is performed on 37 letters of the Urdu language alphabets. Fig. 8 shows the grayscale and segmented images.

Table 2

Average processing time across different resolutions under varying lighting conditions.

Resolution	Average Processing Time (Sec)		
	Bright Light	Moderate Light	Dim Light
640 × 480	2.54	2.44	2.41
800 × 600	2.57	2.45	3.43
1024 × 768	2.61	2.48	2.46
1920 × 1080	2.53	2.40	2.38
2048 × 1540	2.69	2.63	2.47
2560 × 1440	2.98	2.94	2.78
3840 × 2160	3.12	3.09	3.04

The extracted SIFT points are also demonstrated in the Fig. 8.

Upon performing SIFT and determining the sift points, the average duration for each step is evaluated across various lighting conditions and resolution rates, as mentioned in Table 2. From the findings in Table 2, it is evident that the most optimal processing time occurs at a resolution rate of 1920 × 1080 under moderate lighting conditions. This particular resolution rate is deemed suitable for analysis purposes.

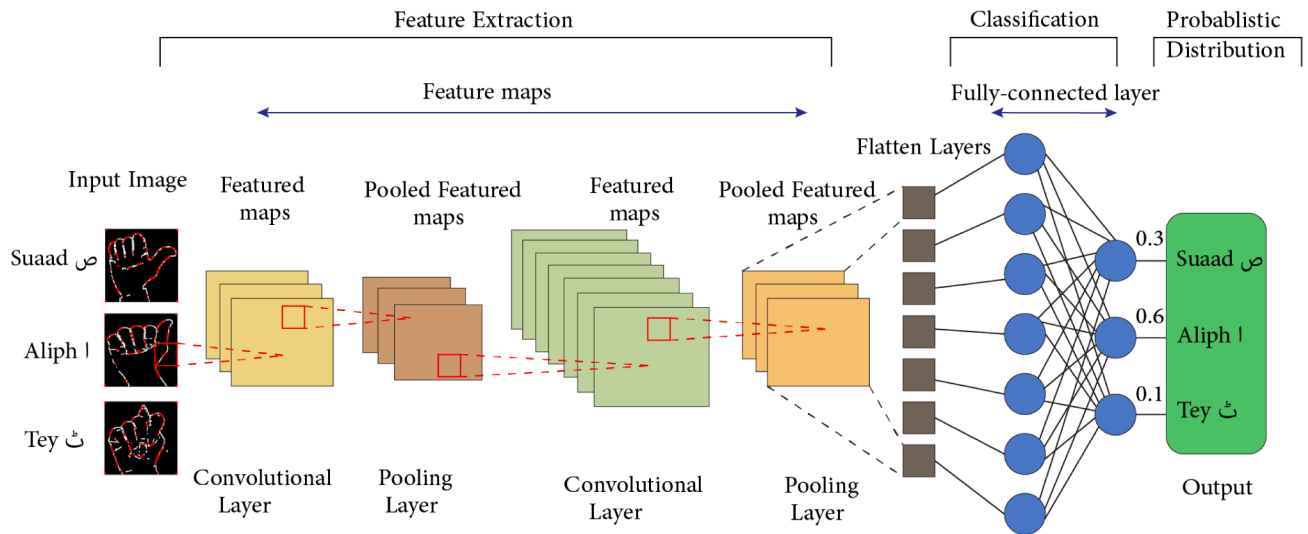


Fig. 9. Proposed Gesture Recognition Classification Architecture using CNN.

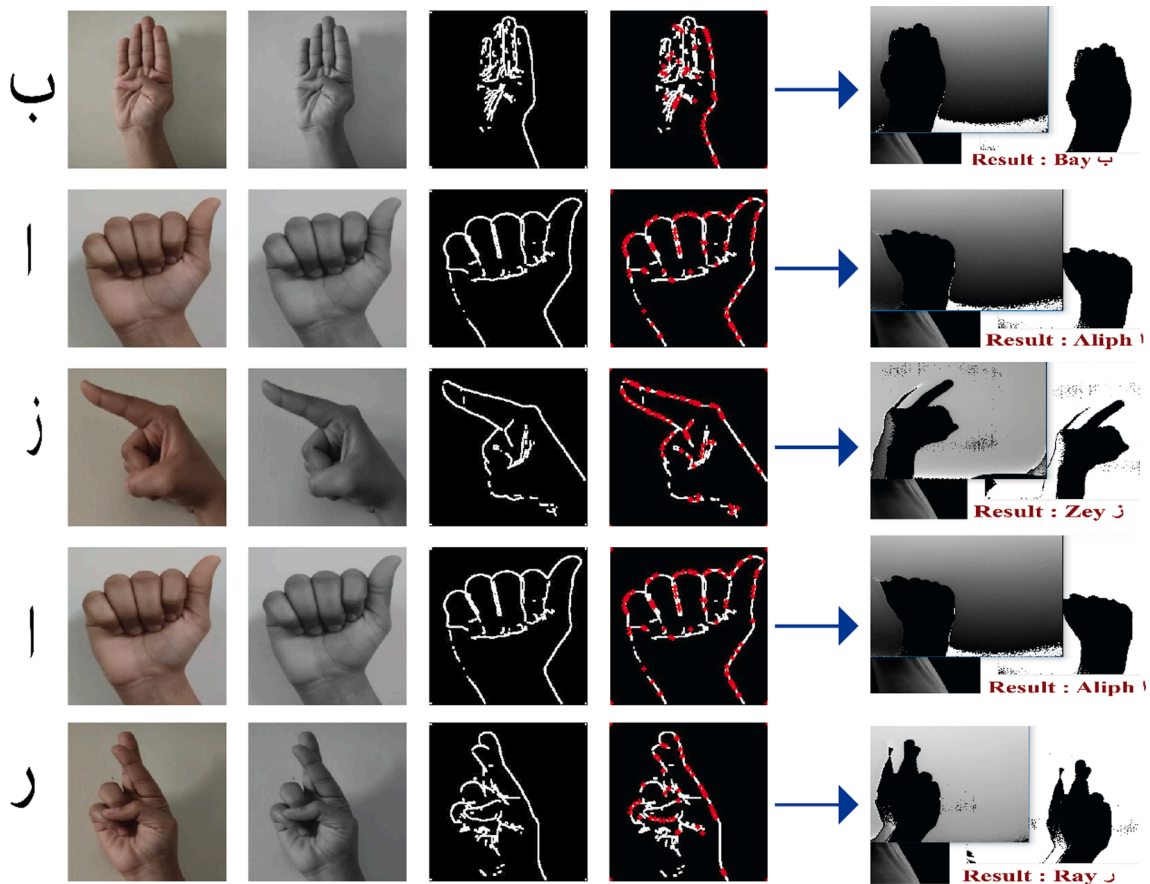


Fig. 10. SIFT-based calculation of key features and significant points in a case study.

After feature points have been calculated and the feature vector has been formulated, the next step is to use these vector points within the proposed CNN framework to identify multiple hand gestures. CNN model training relies heavily on using the right gesture features, so picking them out is an important step. As a result of its profound effect on precision and the way it shapes the network, this is so. While it's true that judiciously choosing features can boost the CNN network's efficiency and cut down on training time, making the wrong choice can have a negative effect on CNN's effectiveness and lead to subpar results.

As shown in Fig. 9, the envisioned CNN framework for PSL recognition works as intended.

For this research, CNN model is designed in a way that plays a crucial role in achieving the best performance. This involves selecting the right number of convolution layers and neurons. However, there are no universal rules for determining the number of layers to choose. In our study, the CNN operates on supervised learning, leveraging labeled data to train the network in recognizing hand gestures accurately. The CNN follows a supervised learning approach and includes specific layers:

Table 3

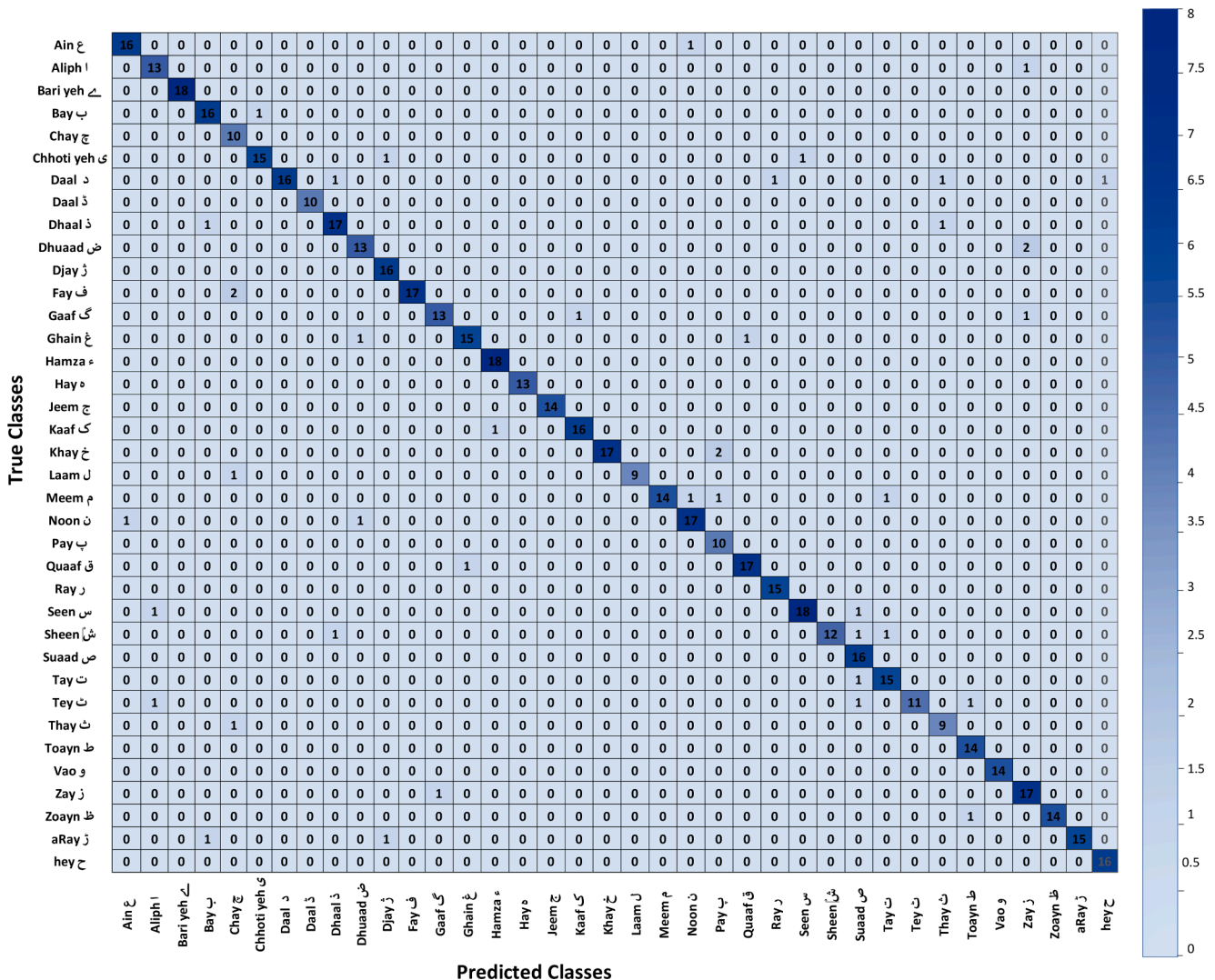
Calculation of feature values for the case study.

Features	Ray ر	Aliph ا	Zay ز	Aliph ا	Bay ب
F1	SD	57.76	68.97	71.05	69.35
	σ	46.25	61.34	53.61	61.97
	x^-	179.83	177.74	156.16	178.87
	M	177.5	179.34	139.75	179.5
	Var	3344.2	4846.83	3819.38	4848.29
F2	SD σ	67.45	98.27	68.53	84.94
	(AD)	47.12	93.44	88.34	73.83
	x^-	164.47	376.42	286.54	453.93
	M	189.53	234.74	324.42	266.87
	Var	5364.52	7384.20	3462.65	6584.54
F3	SD σ	215.34	123.58	157.63	274.46
	(AD)	46.25	89.76	182.98	164.86
	x^-	245.56	267.76	362.66	563.35
	M	209.43	328.83	483.75	195.57
	Var	6674.2	6862.72	6763.35	7368.56
F4	SD σ	178.83	159.82	163.93	175.55
	(AD)	98.76	224.77	264.53	173.64
	x^-	158.83	465.53	524.97	358.56
	M	456.5	134.94	776.83	345.93
	Var	7456.72	2746.97	6736.83	4635.32

input, convolutional, pooling, flatten, fully connected, and output layers for hand gesture recognition. Our parameters and layer configurations were selected through careful experimentation to optimize feature

extraction and classification accuracy. Our CNN design includes five convolution layers, five pooling layers, a flatten layer and two fully connected layer, as well as input and output layers. These components work together to create an effective structure for the CNN. In the case of CNNs, the hidden layer architecture plays a vital role in modeling the learned features. The number of neurons in the hidden layers has a profound impact on the performance of the CNN model. Therefore, the selection of the number of hidden layer neurons is dependent on the CNN's accuracy in initial trials. We have used five hidden layers for our CNN. In our study, we have selected four different features (F1, F2, F3, and F4) as inputs for each layer of the CNN. Each network consists of one convolutional layers, followed by pooling layers, and then fully connected layers. The learned features are hierarchically extracted through the convolutional layers, and the fully connected layers combine these features for classification. The output of the CNN is a vector of classes that represents the recognized hand gestures.

To demonstrate the efficacy of the proposed framework, we first convert the 37 alphabets (call in Urdu language حروف تہجی) to grayscale, segment them, and then calculate the significant points for five letters. In Fig. 10, as a case study, five specific letters, namely “ا”, “ز”, “ب”, “ج”, and “ب” (full form is in URDU language “بازار” and in English language is “Market”). These letters are utilized to demonstrate the process of segmentation, SIFT analysis, and measurement of significant points in hand gestures in Fig. 10.

**Fig. 11.** Confusion matrix for 37 classes of Urdu language alphabets of PSL.

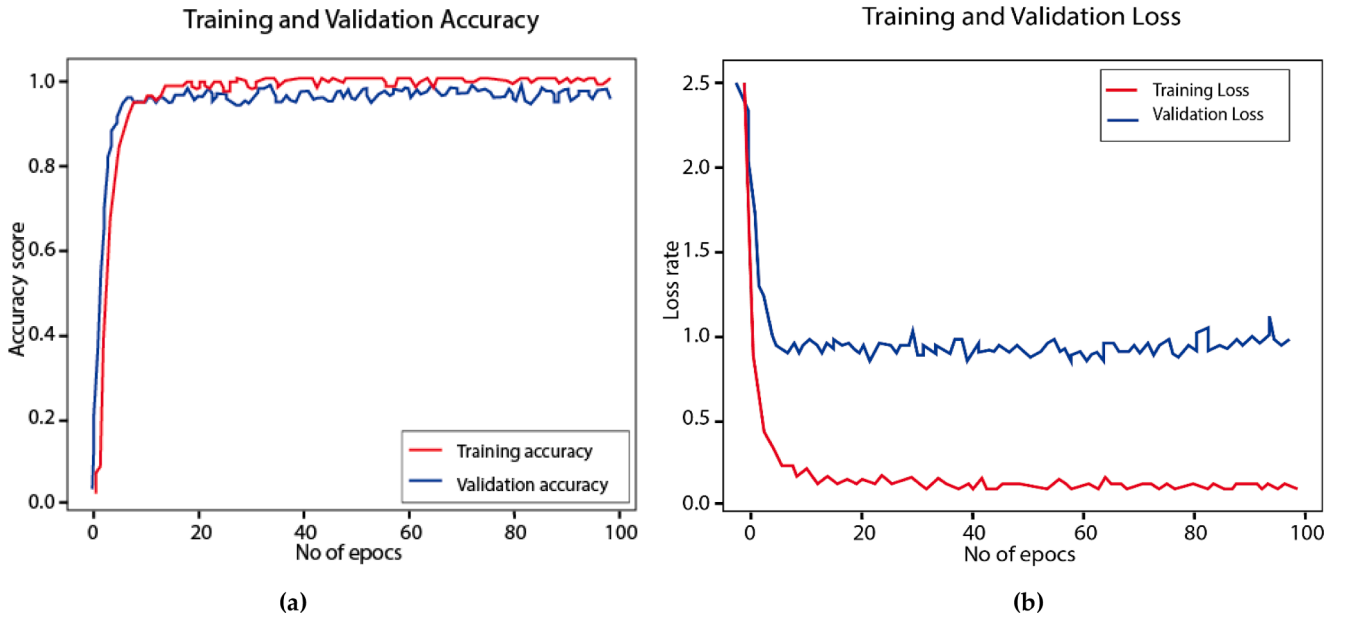


Fig. 12. (a) Training and Validation accuracy (b) Training and Validation loss of the proposed CNN model.

By employing this approach, the framework shows its ability to handle convoluted tasks and accurately capture the complex details of various alphabets and gestures. The Mean ($\bar{x}$), Median (M), Standard Deviation (SD), Average Deviation (σ) and Variance (Var) of each feature is calculated according to Equation (20). We transformed these values into a more summarized representation of the featured SIFT Points. Table 3 represents these values against each feature set.

We built and trained a CNN model using Google Colab, which is an online platform for running Python code. We used Python and TensorFlow to create the CNN model. For classification, we used the Rectified Linear Activation Function (ReLU) in between the convolutional layers (Liu and Liang, 2021 May). The activation function used for fully connected layer is Softmax activation function (Cardarilli et al., 2021 Jul 28). The model was trained with a learning rate of 0.001, a batch size of 32, and 100 epochs (iterations over the dataset). After training, the model achieved an accuracy of 99.65 % on the training data and 98.74 % on the validation data, displaying a good performance by the proposed

model.

In order to evaluate the performance of the proposed model we used confusion matrix for 37 alphabets of PSL. The confusion matrix provides information on the accuracy of gesture recognition. It shows the number of correctly recognized gestures, as well as the number of misclassified gestures and the gestures they are mistaken with. Fig. 11 shows the confusion matrix for PSL.

The Confusion Matrix clearly displays the results obtained from the experiment. By looking at the graph of the confusion matrix, we can easily assess how well the proposed model performed for each class.

Besides accuracy, another method to assess the performance of the proposed system is by analyzing the training loss and validation loss values. These values provide insights into how well the system has adapted to unseen data. By plotting the training and validation losses on a graph, we can visually understand their behavior. Fig. 12(a) indicates the training and validation accuracy to stand at 99.65 % and 98.74 % respectively and Fig. 12(b) shows this representation, indicating that the

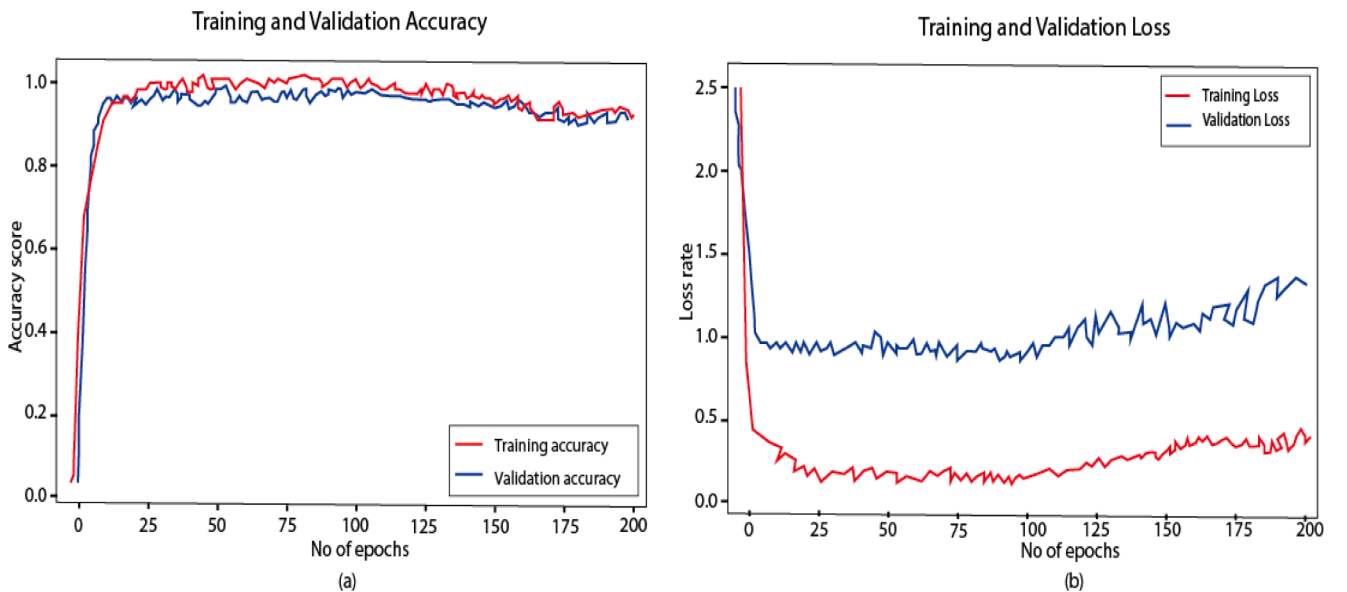


Fig. 13. (a) Training and Validation accuracy with 200 epochs (b) Training and Validation loss of the proposed CNN model with 200 epochs.

Table 4

Comparison of different techniques with proposed method.

Paper	Gesture Count	Technique	Dataset	Accuracy (%)	Error Rate (%)
(Shah et al., 2021)	36	SVM	6633	91.98	8.02
(Imran et al., 2021)	37	SVM	1480	80–90	10–20
(Saleem et al., 2023)	33	CNN	1502	95	5
(Li et al., 2019 Mar)	8	CNN & SVM	complex	98.52	1.48
(Wu, 2020 Apr)	10	DCCNN	720	98.02	1.98
Proposed work	37	SIFT + CNN	2220	98.74	1.26

training loss stands at 0.35 %, while the validation loss is at 1.26 %.

Above Fig. 12 depict the training and validation accuracy as well as loss of the proposed CNN model against each epochs. The accuracy of the model increased as the number of epochs increased. The model achieved a satisfactory level of accuracy 98.74 percent at 100 epochs.

We intentionally designed an experiment with increased epochs to evaluate their impact on our dataset. Interestingly, as the number of training epochs increased, both training and validation accuracy consistently decreased. This decline was also evident in the loss measures. Contrary to expectations, prolonged training resulted in reduced accuracy: 87.76 % for training and an average of 85.34 % for validation. Correspondingly, training and validation losses increased to 0.73 and 1.98, respectively. This unexpected decline suggests overfitting, where the model excessively focused on specific training data nuances, hindering its ability to generalize. To counter this, we explored strategies such as early stopping, regularization, and model simplification to enhance generalization. It is apparent that longer training does not necessarily equate to improved performance, prompting us to reassess our approach for better model learning and adaptation. A visual representation of the effect of increased epochs can be seen in Fig. 13.

In Table 4, we can see a comparison of the proposed system with other existing approaches in terms of accuracy. The results clearly show that the proposed technique performs better than the other methods, which helps us better understand and evaluate the effectiveness of the PSL recognition system.

Fig. 14 is a visual representation of the evaluation of the proposed system versus various state-of-the-art published work. It can be clearly seen in the Fig. 14 below, not only does the proposed model result in the

lowest error rate, but it also yields the highest accuracy among others.

Another comparison of PSL with well-known techniques is presented in Table 5. Here, a more formal results comparison is conducted with some established techniques to validate the approach.

The proposed framework's effectiveness is calculated using performance metrics such as accuracy, precision, recall, and F-score against all targeted classes. The performance of each alphabet in terms of precision, recall, and F-score is depicted in Fig. 15 below.

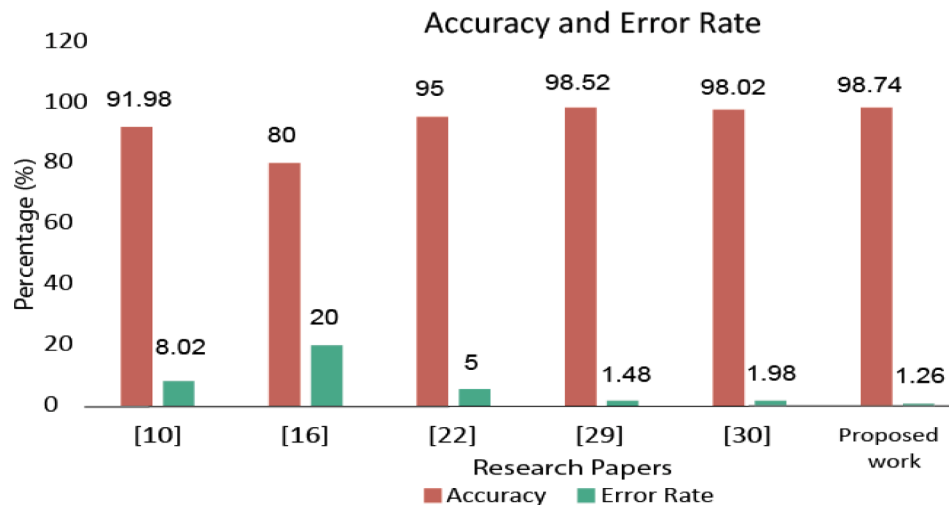
5. Conclusion and future recommendations

Sign language is used as a primary mode of communication by many people who are deaf or hard of hearing. Even though people with hearing impairments have their own language, it can be difficult for those with normal hearing to understand this language. In this research, we explore the feasibility of a vision-based gesture recognition system for PSL. The proposed system is comprised of four main parts: data collection, pre-processing, feature extraction, and classification. The Kinect motion sensor was used to help capture the dataset in a testbed built in the lab. During the process of amassing the dataset, pictures of people's bare hands were taken in a variety of lighting settings and saved in PNG format. The obtained images must then undergo pre-processing in order to standardize their brightness. The luminosity method was used to turn every image into grayscale for this purpose. Images are simplified by being converted to grayscale, making subsequent processes like segmentation less challenging. The thresholding approach is used to perform the segmentation. Selecting too many features requires more resources and computational time, and selecting too few features affects the accuracy of the system, making feature extraction and selection a critical process. The SIFT feature selection and extraction

Table 5

A formal comparison of different well-known techniques with the proposed technique.

Paper	Signs studied	Technique	Accuracy (%)
(Halim and Abbas, 2015)	10	Dynamic Time Wrapping	91
(Hamza and Wali, 2023)	36	3DCNN	93.33
(Malik et al., 2018)	36	Principal Component Analysis and KNN	80
(Raziq and Latif, 2017)	26	ANN	92.5 (but only for 6 alphabets)
Proposed work	37	SIFT + CNN	98.74

**Fig. 14.** Comparison of different techniques with the proposed model in terms of accuracy and error rate.

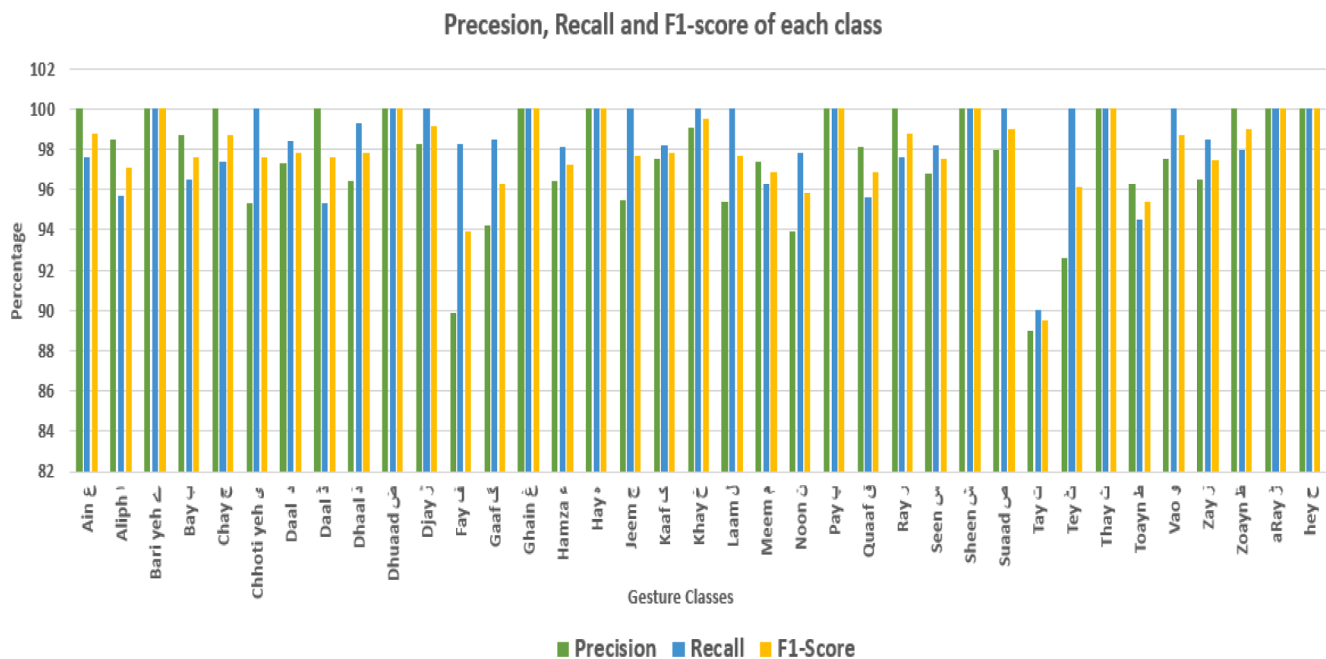


Fig. 15. Precision, Recall and F1-score for each alphabet class for PSL.

method was employed by the system in order to resolve this problem. Margin, hand size and shape, palm center points, and finger position are extracted from input images by the proposed method. After combining these extracted features into a set of feature vector, the SIFT algorithm is applied to the sign images generated by the PSL. Using a descriptor method, these defining features are translated into actionable vector coordinates. The results indicate that the images captured at resolution 1920×1080 rate gives a favourable processing time making it a good resolution for analysis. Increasing resolution rate also increase the complexity hence results in an increased processing time. Our final classification step involved the use of CNN, which have shown promising results in other image recognition tasks. There is a 70:30 split between the Kinect dataset's two subsets, the training set and the testing set. The model achieved a remarkable 99.65 % accuracy after being trained on the training dataset. The model performed only slightly worse on validation data, achieving an accuracy of 98.74 %. The PSL-dataset experienced a loss of 0.35 and 1.26 % during training and validation, respectively.

This research presented provides a solid basis for future expansions and research directions. However, there are a few areas that need more attention and investigation in forthcoming research.

- Comparing PSL to other languages spoken in Pakistan would benefit from a larger, more varied gesture dataset. As a result, it may be difficult to train and evaluate hand gesture recognition frameworks with the data that is currently available. Recognizing different hands in different lighting conditions and with different movements would require a larger dataset.
- Develop real-time video stream hand gesture recognition algorithms. This would make it possible for the system to function in real-world contexts, such as real-time interpreting or interactive software. The need for low-latency performance on devices with limited resources necessitates the investigation of effective software algorithms and methods for optimization.
- Research on the possibility of enhancing hand gesture recognition with additional contextual details. Sign languages use hand gestures and non-manual markers like facial expressions and body postures to communicate. The recognizing system could be improved by adding contextual information.

- The work can be extended by integrating multiple deep learning methods into the hybrid learning model.

Institutional Review Board Statement: The study was conducted according to the guidelines of the Declaration of Helsinki, and approved by the Institutional Review Board of Pir Mehr Ali Shah Arid Agriculture University, Rawalpindi, Pakistan Research Ethics Committee (PMAS-AAUR/R.Eth/63; 13 January 2023).

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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